Stochastic Processes Semester 1

Extension Questions 4 – Solutions

1. (a)

$$\boldsymbol{\pi} = (3/17, 4/17, 6/17, 4/17).$$

(b) For $n \geq 0$, let $\boldsymbol{\mu}^{(n)}$ be the distribution of X_n, i.e. $\mu_i^{(n)} = \mathbf{P}(X_n = i)$. Then

$$\begin{aligned}
\boldsymbol{\mu}^{(1)} &= (1/6, 5/24, 5/12, 5/24) \\
\boldsymbol{\mu}^{(2)} &= (7/36, 35/144, 23/72, 35/144) \\
\boldsymbol{\mu}^{(3)} &= (37/216, 197/864, 161/432, 197/864),
\end{aligned}$$

so that the rate of convergence to the stationary distribution is relatively quick.

2. The stationary distribution $\boldsymbol{\pi}$ associated with this Markov chain is $\boldsymbol{\pi} = (4/7, 3/7)$, and it is immediate that, for all i, j, we have $\lim_{n\to\infty} p_{ij}^{(n)} = \pi_j$.

3. For $i = 0, 1$, let q_i be the probability a policy holder makes i claims in a year. The transition matrix of the Markov chain describing movement amongst the 5 levels, which we label $0, 1, 2, 3, 4$ is

$$P = \begin{pmatrix}
1-q_0 & 0 & q_0 & 0 & 0 \\
1-q_0-q_1 & q_1 & q_0 & 0 & 0 \\
1-q_0-q_1 & 0 & q_1 & q_0 & 0 \\
0 & 1-q_0-q_1 & 0 & q_1 & q_0 \\
0 & 1-q_0-q_1 & 0 & 0 & q_0+q_1
\end{pmatrix}.$$

(a) After one year the vector of the expected numbers at each level is

$$10,000 \times (1-q_0, \quad 0, \quad q_0, \quad 0, \quad 0) = (9048.4, \quad 0, \quad 951.6, \quad 0, \quad 0).$$

After two years the vector of the expected numbers at each level is

$$\begin{aligned}
&10,000\times((1-q_0)^2 + q_0(1-q_0-q_1), \quad 0, \quad q_0(1-q_0+q_1), \quad q_0^2, \quad 0) \\
&\quad =(132.9, \quad 0, \quad 1679.8, \quad 8187.3, \quad 0).
\end{aligned}$$

(b) Let $\boldsymbol{\pi}$ be he stationary distribution of the chain. We see that

$$\begin{aligned}
\pi_0 &= \frac{(1+q_0-q_1)(1-q_0-q_1)^2}{(1-q_1)^3} \\
\pi_1 &= \frac{q_0^2(1-q_0-q_1)}{(1-q_1)^3} \\
\pi_2 &= \frac{q_0(1-q_0-q_1)}{(1-q_1)^2} \\
\pi_3 &= \frac{q_0^2(1-q_0-q_1)}{(1-q_1)^3} \\
\pi_4 &= \frac{q_0^3}{(1-q_1)^3}
\end{aligned}$$

So the expected numbers at each discount level once stability has been achieved are given by

$$10,000\boldsymbol{\pi} = (0.5, \quad 50.9, \quad 51.2, \quad 50.9, \quad 9846.5).$$

4. (d) It is clear by symmetry that the stationary distribution is given by $\boldsymbol{\pi} = (1/k,\, 1/k,\, 1/k,\, \ldots,\, 1/k)$. Further $\lim_{n\to\infty} P^n = E$ as required.

   (e)

$$f_{ii}(n) = \begin{cases} 0 & \text{when } n = 1 \\ \left(\frac{k-2}{k-1}\right)^{n-2} \frac{1}{k-1} & \text{when } n \geq 2 \end{cases}$$

   So we can find that the mean number of days between successive wearings of pair of socks i is k

5. (a) There are 6 possible states:

$$(1,2,3), (1,3,2), (2,1,3), (2,3,1), (3,1,2), (3,2,1).$$

   We label them as states 1 to 6, for short. So the transition matrix is

$$P = \begin{pmatrix} p_1 & 0 & p_2 & 0 & p_3 & 0 \\ 0 & p_1 & p_2 & 0 & p_3 & 0 \\ p_1 & 0 & p_2 & 0 & 0 & p_3 \\ p_1 & 0 & 0 & p_2 & 0 & p_3 \\ 0 & p_1 & 0 & p_2 & p_3 & 0 \\ 0 & p_1 & 0 & p_2 & 0 & p_3 \end{pmatrix}$$

   We can see that all states intercommunicate, so there is only one stationary distribution. To find it, we have again to solve the system of equations $\pi P = \pi$ together with normalization $\sum \pi_i = 1$.
   The solution is

$$\pi = \left(\frac{p_1 p_2}{1-p_1}, \frac{p_1 p_3}{1-p_1}, \frac{p_2 p_1}{1-p_2}, \frac{p_2 p_3}{1-p_2}, \frac{p_3 p_1}{1-p_3}, \frac{p_3 p_2}{1-p_3} \right)$$

   which has a nice symmetry.
   Since this Markov chain is aperiodic, the limiting distribution coincides with the stationary, so

$$\alpha_n \to \pi_1 = \frac{p_1 p_2}{1-p_1}, \quad \text{as} \quad n \to \infty.$$

   (b) We have $\pi_3 + \pi_4 = p_2$, the two states with 2 on top. So the long-run proportion of time is p_2.